The New Economics of Multisided Platforms:

A Guide to the Vocabulary

David S. Evans and Richard Schmalensee*

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Abstract

The new economics of multisided platforms involves a number of concepts that are familiar to economists as well as some new ones. This glossary, which is drawn from our book *Matchmakers: The New Economics of Multisided Platforms*, is an attempt to put together the main concepts and to provide short definitions for them. We believe this common vocabulary, which we're sure others will refine and evolve, will be helpful for research and the exchange of ideas going forward and may serve as a quick primer for those who are new to the field, including students.

* Evans is Chairman, Global Economics Group; Visiting Professor, University College London; and Lecturer, University of Chicago Law School. Schmalensee is Howard W. Johnson Professor of Management and Economics Emeritus at MIT and Dean Emeritus of the MIT Sloan School of Management. Email correspondence to devans@globaleconomicsgroup.com. This glossary is drawn from

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Matchmakers: The New Economics of Multisided Platforms. We thank Harvard Business Review Press for allowing us to reproduce it.

Guide to the Vocabulary of the New Economics of Multisided Platforms

The new economics of multisided platforms involves a number of concepts that are familiar to economists as well as some new ones. This glossary, which is drawn from our book *Matchmakers: The New Economics of Multisided Platforms*, is an attempt to put together the main concepts and to provide short definitions for them.[1] We believe this common vocabulary, which we're sure others will refine and evolve, will be helpful for research and the exchange of ideas going forward and may serve as a quick primer for those who are new to the field, including students.

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access fee: The price that customers are charged for obtaining access to the platform. For example, credit card issuers sometimes charge people an annual fee for using the card, thereby obtaining access to the merchants that accept these cards for payment.

anchor tenant: A significant customer that helps attract other customers to the platform where the customers could be the same type as the anchor tenant, or the customers could be a different type of customer that is attracted to the anchor tenant. For example, department stores and big-box retailers are anchor tenants at shopping malls; they attract shoppers and they also persuade smaller retailers to take space at the mall in anticipation that these anchor stores will attract traffic.

behavioral externality: Actions taken by platform participants that directly affect other participants positively or negatively. They are different from network externalities because they don't involve the number of participants but rather the behavior of those participants. For example, an app developer imposes costs on end users when it inserts malware into its app.

critical mass: The minimum sets of participants for the platform sides that are necessary for the platform to ignite—that is, have self-sustaining growth. For a two-sided platform, critical mass isn't typically a single pair of numbers for each type of participant, but rather there is a range of numbers that could provide critical mass. For example, a B2B

[1] David S. Evans and Richard Schmalensee, *Matchmakers: The New Economics of Multisided Platforms* (Boston: Harvard Business Review Press, 2016). Available here.

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exchange has to have enough buyers and enough sellers to interest either side; once it has enough, more will join, while if it doesn't, it will lose the ones it has attracted.

death spiral: The reinforcing loss of participants on either side of a platform that results when a platform loses a critical mass of customers and eventually collapses. For example, dead malls result from a death spiral in which the mall loses shopper traffic, which results in a loss of retailers, which results in a further loss of traffic, until eventually no retailers want to locate at the mall and no shoppers want to come.

direct network effect: The impact of the addition of another participant to a network on other participants in the same group. A positive network effect occurs when an additional participant makes other participants of the same sort better off because they can reach and interact with more participants. For example, suppose people both make and receive phone calls to roughly the same degree. An additional person increases the size of the network and increases the number of people all other members can connect with. A negative direct network effect occurs when additional participants make other participants in the same group worse off, perhaps because of congestion or competition; for example, when an additional man goes to a singles bar, it increases competition for other men and may also make it too crowded to easily mingle.

ecosystem: The businesses, institutions, and other environmental factors that affect the value, positively or negatively, that a platform can generate for the participants on the platform. For example, the value of a shopping mall to retailers is greater if there is easy road access to it and if it is located farther away from competing shopping malls, single-standing department stores, and shopping streets.

edge provider: An online business that provides services or content by connecting to end users through an Internet Service Provider (ISP). For example, Pandora provides music over the Internet by connecting to people who listen to it over a fixed ISP such as a cable television system that provides broadband or a mobile ISP such as mobile network operator.

externality: A benefit or cost that one participant imposes on another participant without direct monetary compensation. An externality could arise from a network effect or from a behavioral externality. See the examples for behavioral externality, direct network effects, and indirect network effects.

foundational platform: A multisided platform that provides core services to other multisided platforms and is therefore a "platform for platforms." These include Internet Service Providers (ISPs), which connect edge providers and end users. For example, Comcast makes it possible for end users to connect over the Internet to Google's search engine. Foundational platforms also include computer operating systems, or invisible engines, which provide a standard platform for app developers and end users; for example, Android provides an operating system that enables app developers to provide apps to end users and for end users to use those apps.

friction: Costs or other impediments that impede mutually advantageous interactions and exchanges. For example, before there were online reservation services, people who wanted to make dinner reservations had to identify restaurants, get their phone numbers, call them, see if they had a table, possibly leave a message, and try other restaurants if their first choice wasn't available; meanwhile, restaurants had to have someone answering the phone and recording reservations in a notebook.

governance system: Rules that prohibit bad behavior (that is, negative *behavioral externalities*) by platform participants and mechanisms for enforcing those rules, including methods of detection and punishment. Marketplaces such as eBay, for example, have rules that prohibit sellers from engaging in various activities, such as posting misleading content, that can harm buyers and that prohibit buyers from engaging in various activities, such as not paying for items they have agreed to buy, that can harm sellers.

ignition: When a multisided platform achieves critical mass and starts a process of self-sustaining growth. For example, once M-PESA had enough people with accounts who wanted to send or receive, and enough cash-in/cash-out agents who sold e-money to senders or who redeemed e-money from receivers, it started to grow very quickly.

implosion: When a multisided platform can't achieve or maintain critical mass, it starts losing customers and eventually goes into a death spiral. For example, many mobile-money platforms have been unable to get enough subscribers or agents to ignite and have instead withered away.

indirect network effect: The impact of the addition of one type of participant to a network on another type of participant. A positive indirect network effect arises when an additional participant of one type increases the value that participants of another type get. For example, an additional restaurant on OpenTable increases the value to diners, who now have an additional place to consider and at which to make reservations. A negative indirect network effect arises when an additional participant of one type decreases the value to participants of the other type. For example, more radio ads reduce the value that radio listeners get.

Internet service provider (ISP): An entity the enables end users and edge providers to connect to each other through the Internet. Fixed ISPs have physical wires or cables into households, apartment buildings, and businesses to make connections; people or businesses may then operate wireless networks at those physical locations. Mobile ISPs connect to people with mobile devices over the wireless spectrum using cell towers. For example, Comcast is a fixed ISP operating through a cable system, while T-Mobile is a wireless ISP.

Invisible engine: A computer operating system that provides services through "application programming interfaces" (APIs) that apps can use and that end users can use to run those apps. Invisible engines generally operate computer hardware, including the central processing chip, and make that functionality available to apps and users.

Microsoft Windows is an example of an invisible engine for personal computers. Facebook is also an invisible engine because it provides APIs for app developers who write apps that people can use on Facebook. Invisible engines are sometimes called "software platforms."

marquee customer. See "anchor tenant."

matchmaker: A business that helps two or more different kinds of customers find each other and engage in mutually beneficial interactions. Matchmaking does not involve literally finding perfect matches for people—like the old village matchmaker would try to do for a potential marriage—but rather finding good trading parties. A payment card network, for example, helps retailers and consumers get together and transact by using the same, agreed-on payment method. Also see "multisided platforms," which is another name for matchmakers.

money side: For two-sided platforms, a group of customers that provides all or virtually all of the profits earned by the platform. For multisided platforms, one or more groups could not contribute any profits (see "subsidy-side"), while two or more groups together could provide all or virtually all of the profits. For example, most newspapers make all of their profit from advertising and subsidize readers by providing them with content.

multihoming: When platform participants use two or more similar platforms or could easily do so. For example, many consumers carry several different payment cards and select one of them to pay when they go to the store.

multisided platform: A business that operates a physical or virtual place (a platform) to help two or more different groups find each other and interact. The different groups are called "sides" of the platform. For example, Facebook operates a virtual place where friends can send and receive messages, where advertisers can reach users, and where people can use apps and app developers can provide those apps.

network effect. See "direct network effect" and "indirect network effect."

operating system: See "invisible engine."

pioneering platform: A multisided platform that is the first, or one of the first, to identify a friction and create a matchmaker to attempt to solve that friction. A pioneering platform, unlike later attempts to solve the same friction, usually has to be one of the first to solve the pricing, chicken-and-egg, and design issues necessary to solve the friction and ignite a platform. For example, OpenTable was one of the first online restaurant-reservation matchmakers and had to solve the key business issues itself with no guidance from the experience of other firms.

pricing level: The level of prices that a multisided platform charges to both sides. When the prices to the two sides are fees for transactions, the total price level is the total amount paid by both sides for a transaction. For example, exchanges have set fees for liquidity providers (usually negative) and liquidity takers (usually positive), and the total price level is the total fees to the exchange for a trade, the sum of the fee received by the liquidity provider, and the fee paid by the liquidity taker.

pricing structure: The distribution of the price or revenue contributed by the different sides of a multisided platform. For example, for fleet cards for paying truck stops on behalf of truck fleets, fleets pay about 25 percent of the total price and the truck stops pay about 75 percent. One side can pay more than 100 percent when there are negative prices to the other side or sides. Restaurants pay more than 100 percent of fees for OpenTable since diners pay no fees and are paid reward points.

screening device: A method for limiting participation on the platform to participants on each side that are attractive to participants on the other side. Common screening devices are "exclusionary vibes," which provide signals that only certain kinds of participants (such as lower-income people for a mall, or Jewish people for a dating site) should join the platform; and "exclusionary amenities," which provide value only to the type of participants that the platform seeks to attract (such as articles on fly fishing for a magazine that wants to attract purchasers of fly-fishing-related goods and services).

self-supply: For platforms that connect consumers with providers of goods and services, the platform supplies some of those goods and services itself. It may do this as a tactic during its start-up to help secure ignition, and it may do this longer term simply because it has some expertise in doing so and can make money that way.

signaling device. An indicator that participants belong to the same platform and could therefore profitably interact. Payment card systems, for example, use logos on cards and at merchant terminals to inform cardholders that they can use their card at that terminal and for merchants to inform cardholders of this fact.

single-homing: When platform participants standardize on and use only a single platform. For example, most people have a single wired broadband provider at home.

subsidy-side: A group of customers who do not cover their costs of participating on a platform. A multisided platform could have more than one group of subsidized customers, so long as there is one group that functions as the money side. Most consumers of ad-supported media, for example, pay little if anything and are provided content of significant value.

transaction cost: The economists' term for frictions; see "friction."

turbocharged matchmaker: A matchmaker, or multisided platform, that benefits from a significant combination of powerful computer chips, the Internet, the web, broadband communications, programming languages and operating systems, and the Cloud. Airbnb, for example, benefits from all of these technologies.

two-sided market: The original name used to refer to industries that had "two-sided platforms." Two-sidedness is a characteristic of businesses, not always of industries, however.

two-step strategy: An ignition strategy in which the platform secures significant participation by one group and then secures participation by the other group by offering them access to the first group. Advertising-supported media typically do this by securing eyeballs first and then selling access to them to advertisers.

usage externality: The benefit that one party receives as a result of engaging in an exchange with another party. Two people benefit from being able to use a matchmaker to engage in a transaction, even if they can't connect with any other people. All multisided platforms have indirect network externalities as well as usage externalities.

usage fee: A fee paid by one or more participants on a multisided platform for interacting with another participant. For example, OpenTable charges restaurants a $1 per person reservation fee.

zigzag strategy: An ignition strategy in which the platform adopts tactics to increase participation of the different sides simultaneously. It is called zigzag because it imagines that the platform tries to push one group's participation up, then works on another group, and continues this process to secure positive feedbacks. In practice, it may push on all groups at the same time but vary the relative efforts devoted to them.